

Abdul Wahab Ahmed Khan

Software Engineer — Backend Systems & ML Infrastructure

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[LinkedIn](#) | [GitHub](#) | [LeetCode \(1000+\)](#) | [AtCoder \(Cyan, 1200\)](#)

[Codeforces \(Specialist 1477\)](#) | [CodeChef \(3-Star, 1680\)](#) | [Kaggle](#) | [Portfolio](#)

Professional Experience

QF Network

Karachi, Pakistan

Full Stack Developer

Mar 2025 – Present

- Optimized **p99 API latency by 40%** by refactoring Prisma schema and implementing composite **B-Tree indexing** in PostgreSQL.
- Developed a **semantic search engine** using **pgvector (HNSW indexing)** for high-dimensional retrieval across **10,000+ SKU** catalogs.
- Architected an **LLM pipeline** in Next.js, reducing merchant onboarding time by **95% (hours to 5 mins)** via automated context and rate-limit management.
- Designed a **distributed worker system** using **BullMQ and Redis**, eliminating thread-blocking operations for asynchronous reward and webhook delivery.

Applied Machine Learning

Stanford RNA 3D Folding | Rank: 55/2,226 (Top 2.5%)

Scientific ML

- Engineered Transformer/CNN models with custom spatial embeddings; optimized GPU data pipelines for memory-constrained biological datasets.

Thermophysical Property Prediction | Rank: 62/1,176 (Top 5%)

Chemical Informatics

- Developed regression models using SMILES-derived graph descriptors to predict molecular melting points and improve convergence stability.

NVIDIA Nemotron Reasoning Challenge | Rank: 69/432 (Top 16%) — LLM Fine-Tuning

- Performed supervised fine-tuning (SFT) on foundational LLMs using reasoning datasets, improving multi-step logical consistency through optimized gradient updates.

Competitive Programming

Algorithmic Problem Solving & Competitive Programming

- AtCoder**: Ranked **67th** globally out of 10,000+ participants in ABC 453; reached **Cyan (1200)** rating in a single debut match (Top 1% globally).
- Codeforces**: Specialist (Max Rating 1477); regular Div.2 participant specializing in graph algorithms, dynamic programming, and number theory.
- LeetCode**: Solved 1000+ problems across advanced data structures and optimization techniques (Top 5% globally).
- CodeChef**: 3-Star (1680 rating); achieved +450 rating increase in a single contest (Starters 234), ranking 7th in Pakistan.

Technical Skills

Languages: C++ , Python, Go, TypeScript, Java, SQL

ML Infrastructure: JAX, PyTorch, XLA

Backend/Systems: Node.js, Spring Boot, BullMQ, Redis

DevOps & Data: PostgreSQL, Docker, Google Cloud, CI/CD Pipelines

Systems Engineering Projects

High-Concurrency AI Orchestrator | Go, Docker, Distributed Systems

[GitHub](#)

- Engineered a high-throughput distributed orchestration engine in Go, implementing worker pools and channel-based concurrency to safely manage parallel LLM API workloads without goroutine leaks or blocking execution.

Distributed Media Streaming Platform | Spring Boot, PostgreSQL, Azure

[GitHub](#)

- Designed scalable media backend with stateless JWT authentication and optimized payload chunking for low-latency streaming across distributed client environments.

Education

Virtual University of Pakistan

Expected 2029

Bachelor of Science in Computer Science

Karachi, Pakistan

- Coursework**: Data Structures & Algorithms, Discrete Math, Systems Architecture, Calculus.